

Yi-Wen Hsiao

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Personal Profile

A PhD candidate at National Taiwan University, undertaking biomedical big data projects. I have been trained in Biostatistics, Bioinformatics, and Biotechnology, with experimental and computational skills developed from the extensive research experience. My current research interests involve highly dimensional data analysis, artificial intelligence/machine learning, and their application in the biological field. Primarily looking for a postdoctoral job.

Education

Institute of Epidemiology and Preventive Medicine, College of Public Health, National Taiwan University

Taipei, Taiwan

PhD candidate in Biostatistics

Sep 2019 - Jul 2023

- Ministry of Science and Technology (MOST)'s Outstanding Doctoral Scholarship Pilot Program (\$2,4000 per year)
- Passed oral thesis defense with distinction (Mar 2022)
- **PhD Thesis Title:** Development of Prognostic Models for Patients with Different Cancer Types by Using High-throughput Genomic Data
- **Courses:** Mathematical Statistics, Statistical Analysis for Repeated Measurements, Statistical and Machine Learning

School of Natural and Environmental Sciences, Newcastle University

Newcastle upon Tyne, UK

MSc in Biotechnology

Sep 2014 - Sep 2015

- Graduated with merit
- **Dissertation Title:** Ribosome engineering to improve antimicrobial production

College of Engineering, National Chung Cheng University

Chiayi, Taiwan

MSc in Chemical Engineering

Sep 2010 - Jan 2013

- Graduated with GPA 4.24/5.0
- **Dissertation Title:** Differentiation of Dopaminergic Neurons from Bone Marrow-Derived Mesenchymal Stem Cells Modified by DNA Target Methylation

College of Engineering, National Chung Cheng University

Chiayi, Taiwan

Bachelor in Chemical Engineering

Sep 2010 - Jan 2013

- Graduated with GPA 3.75/5.0

Work Experience

Bioinformatics and Biostatistics Core Lab, Center of Genomic and Precision Medicine, National Taiwan University

Taipei, Taiwan

Research Assistant

Mar 2016 - Jul 2019

- Analyzed different types of microarray and sequencing data, including WGS/WES, RNA-seq (mRNA, miRNA, circRNA and lncRNA), ATAC-seq and 16S and ITS rRNA Sequencing, across different species.
- Built customized NGS data analysis pipelines and provided statistical results, data visualization, and basic biological interpretation
- **Technical Skills:** R, Python, Linux.

Research Project

Visiting Research Student

Melbourne, Australia

Melbourne Integrative Genomics, School of Mathematics and Statistics, University of Melbourne

Jul 2022 - Jun 2023

- Graduate Students Study Abroad Program provided by Taiwanese Ministry of Science and Technology (MOST) (\$45,000 a year)
- Studying statistical approaches for integrating biological data generated from different sequencing technologies.
- Generalizing the analytical pipeline for disease-based studies.
- **Technical Skills:** R, Python, Overleaf, LaTeX.

Publications

Hsiao, Y. W., Wang, L., and Lu, T. P. (2022). ceRNAR: An R package for identification and analysis of ceRNA-miRNA triplets. PLoS Computational Biology, 18(9), e1010497.

Hsiao, Y. W., and Lu, T. P. (2021). Race-specific genetic profiles of homologous recombination deficiency in multiple cancers. Journal of Personalized Medicine, 11(12), 1287.

Hsiao, Y. W., Tao, C. L., Chuang, E. Y., and Lu, T. P. (2021). A risk prediction model of gene signatures in ovarian cancer through bagging of GA-XGBoost models. Journal of advanced research, 30, 113-122.

Juang, J.M.J., Lu, T.P., Su, M.W., Lin, C.W., Yang, J.H., Chu, H.W., Chen, C.H., **Hsiao, Y.W.**, Lee, C.Y., Chiang, L.M. and Yu, Q.Y. (2021) Rare variants discovery by extensive whole-genome sequencing of the Han Chinese population in Taiwan: Applications to cardiovascular medicine. *Journal of Advanced Research*, 30, pp.147-158.

Hsiao, Y. W., Chiu, L. T., Chen, C. H., Shih, W. L., and Lu, T. P. (2019). Tumor-infiltrating leukocyte composition and prognostic power in hepatitis B-and hepatitis C-related hepatocellular carcinomas. *Genes*, 10(8), 630.

Hsiao, Y. W., and Lu, T. P. (2019). Text-mining in cancer research may help identify effective treatments. *Translational Lung Cancer Research*, 8(Suppl 4), S460.

Chao, K.H., **Hsiao, Y.W.**, Lee, Y.F., Lee, C.Y., Lai, L.C., Tsai, M.H., Lu, T.P. and Chuang, E.Y.. (2019) RNASeqR: an R package for automated two-group RNA-Seq analysis workflow. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 18(5), 2023-2031.

Lai, L.C., Sun, Q.L., Chen, Y.A., **Hsiao, Y.W.**, Lu, T.P., Tsai, M.H., Zhu, L., Chuang, E.Y. and Fang, W. (2019) Using proteomic profiling to characterize protein signatures of different thymoma subtypes. *BMC cancer*, 19, 1-8.

Lee, C.H., Kao, Y.C., Lee, W.R., **Hsiao, Y.W.**, Lu, T.P., Chu, C.Y., Lin, Y.J., Huang, H.Y., Hsieh, T.H., Liu, Y.R. and Liang, C.W. (2019) Clinicopathologic characterization of GREB1-rearranged uterine sarcomas with variable sex-cord differentiation. *The American journal of surgical pathology*, 43(7), 928-942.

Tseng, T.C., Huang, D.Y., Lai, L.C., Hwai, H., **Hsiao, Y.W.**, Jhou, J.P., Chuang, E.Y. and Tzeng, S.J. (2018) Dual immuno-renal targeting of 7-benzylidenenaltrexone alleviates lupus nephritis via FcγRIIB and HO-1. *Journal of Molecular Medicine*, 96, 413-425.

Tsai, C. H., Tzeng, S. F., Hsieh, S. C., Yang, Y. C., **Hsiao, Y. W.**, Tsai, M. H., and Hsiao, P. W. (2017). A standardized herbal extract mitigates tumor inflammation and augments chemotherapy effect of docetaxel in prostate cancer. *Scientific Reports*, 7(1), 1-14.

Conferences

2022	Australian Mathematical Sciences Institute (AMSI) Bioinfo Summer , The ceRNAR package to identify ceRNA events in pan-cancer studies (Poster + Fast forward talk)	Melbourne, Australia
2022	Australian Bioinformatics and Computational Biology Society , A computational framework to detect ceRNA cross-talks in pan-cancer studies (Poster)	Melbourne, Australia
2022	Australian Bioinformatics and Computational Biology Student Society , A computational framework to detect ceRNA cross-talks in pan-cancer studies (Oral)	Melbourne, Australia
2015	Taiwan Public Health Association TPHA , A genomewide association study of Brugada Syndrome in Taiwan (group work) (Oral)	Taipei, Taiwan
2015	International Conference on Intelligent Biology and Medicine ICIBM , The Comparisons of Prognostic Power and Expression Level of Tumor Infiltrating Leukocytes in Hepatitis B- and Hepatitis C-related Hepatocellular Carcinomas (Oral)	Ohio, USA

Teaching Assistant

2021	Applied Biostatistics (B) , Fall	Taiwan
2021	Applied Biostatistics (A) , Fall	Taiwan
2021	Biostatistics and Epidemiology , Spring	Taiwan
2021	Introduction and Application of Computational Biology Methods , Spring	Taiwan
2020	Applied Biostatistics (B) , Fall	Taiwan

Skills

Programming R, Python.

Miscellaneous Linux, Shell (Bash/Zsh), $\LaTeX$ (Overleaf/R Markdown), Microsoft Office, Git.

Languages

English Professional proficiency

Mandarin Native proficiency

Interests

Travel I love traveling. Since 2015, I have been to over ten countries/regions.

Sports I usually work out in the gym. I enjoy various activities, including hiking, boxing, cycling, and paddling.